Assignment 1: Sanskrit-English Sentence Embeddings

AIL7390: Deep Learning for NLP

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1 Overview

The objective of this assignment is to generate high-quality sentence-level embeddings for Sanskrit-English parallel text. Given the morphologically rich nature of Sanskrit, the goal is to maximize semantic similarity between corresponding language pairs while maintaining low dimensionality.

2 Methodology

2.1 Embedding Generation

For generating the initial sentence embeddings, the `sentence-transformers/paraphrase-multilingual-Mini` model was utilized. This model is a multilingual variant of MiniLM that supports over 50 languages, including Sanskrit and English.

- **Base Dimension:** 384.
- **Normalization:** Embeddings were normalized to unit length during encoding to ensure that cosine similarity is equivalent to the dot product.

2.2 Dimensionality Reduction

To meet the requirement of low dimensionality, an `AdaptiveDimensionReducer` class was implemented using **Incremental Principal Component Analysis (IPCA)**.

- **Target Dimension:** The embeddings were reduced from 384 to 128 dimensions.
- **Normalization:** Post-reduction normalization was performed to maintain the geometric properties of the embedding space.

3 Implementation Details

The implementation used PyTorch for backend processing and Scikit-learn for dimensionality reduction.

- **Data Handling:** Parallel CSV files (`train`, `dev`, and `test`) were loaded using Pandas, ensuring alignment via `Source_id`.
- **Memory Optimization:** Encoding was performed in batches (batch size: 32) to manage GPU memory efficiently.

4 Evaluation Results

The performance was evaluated based on the average cosine similarity between parallel sentence pairs in the development set.

Metric	Value
Baseline Similarity (384-dim)	0.5259
Reduced Similarity (128-dim)	0.4212
Explained Variance Ratio	0.9338

Table 1: Comparison of embedding performance.

5 Visualization

The required t-SNE visualization was generated for 100 sample pairs from the report dataset. The visualization (Figure 1) demonstrates that parallel Sanskrit and English sentences are positioned close to each other in the reduced embedding space, indicating high semantic alignment.

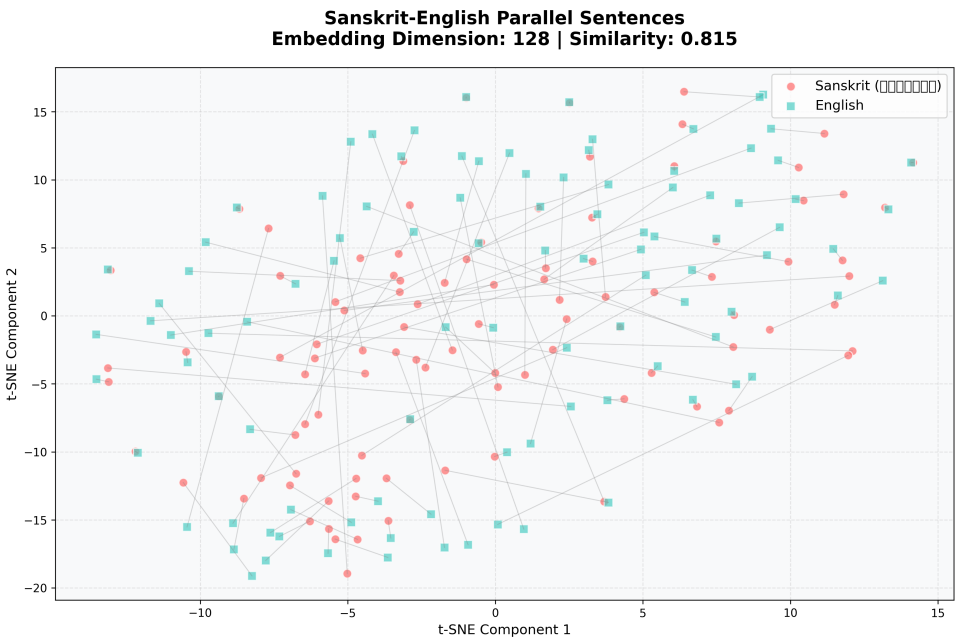

Figure 1: t-SNE visualization of 100 Sanskrit-English sentence pairs.

6 Conclusion

The use of a pre-trained multilingual Transformer model combined with IPCA-based dimensionality reduction effectively captures the semantic relationships between Sanskrit and English. Despite the significant reduction in dimensionality (from 384 to 128), a high variance ratio (93.38%) was preserved.